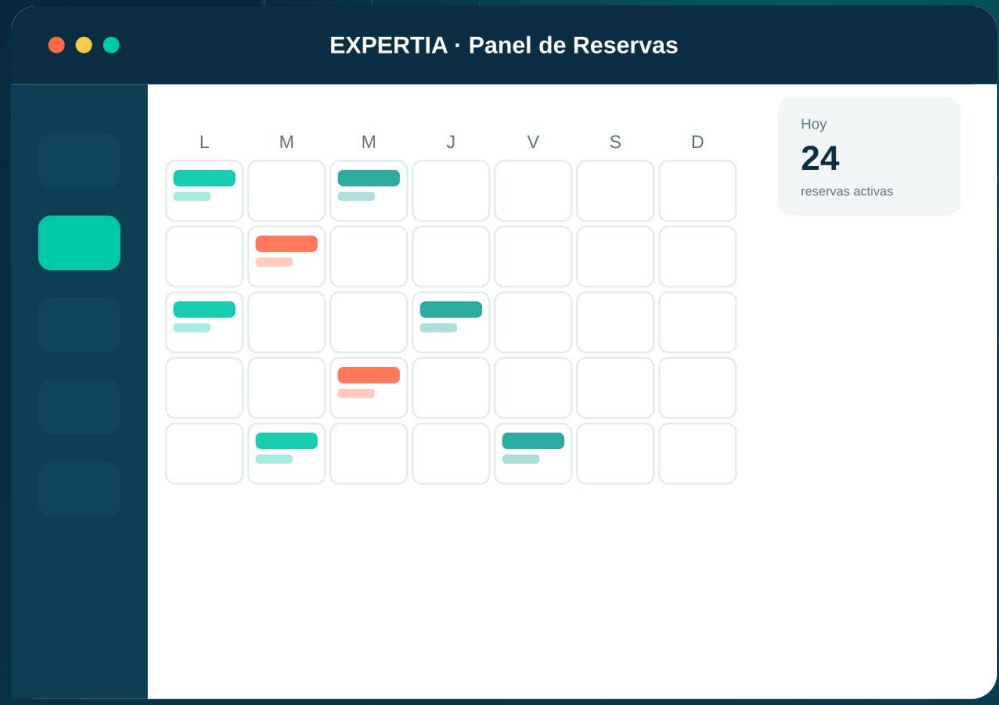
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FINAL PRESENTATION · PHASE 3

EXPERTIA

Business Management Platform

Multi-tenant SaaS booking and management platform, easy to set up for any service business, with an AI scheduling assistant.



Team Maximiliano Díaz · Genesis Cáceres · Alberto Rincón

Faculty advisor Félix Cifuentes

Computer Engineering · Duoc UC · November 2026

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TEAM

Project team

MD

Maximiliano Díaz

PROJECT MANAGER

RESPONSIBILITIES

- Phased strategy and sprint plan
- Team and Jira organization
- Repository setup
- Documentation in Confluence
- Progress and risk tracking

GC

Genesis Cáceres

SENIOR FUNCTIONAL ANALYST

RESPONSIBILITIES

- Requirements and user stories
- Acceptance criteria per sprint
- Functional validation of each story
- Quality and coverage strategy
- Functional documentation

AR

Alberto Rincón

FULL STACK DEVELOPER

RESPONSIBILITIES

- Angular 18 frontend
- Django REST backend
- Docker environment and CI pipeline
- Gemini conversational assistant
- Functional mockup and prototype

PROJECT DESCRIPTION

From problem to solution

Problem or pain point

Small and mid-sized service businesses

- Lost customers due to manual booking
- Scheduling errors and double bookings
- No operational visibility of the business

The market forces a trade-off: no competitor (AgendaPro, Calendly, Booksy, Fresha) combines versatility, simplicity and transparent pricing.

Proposed solution

Expertia multi-tenant SaaS

Versatile

One Service ↔ Resource model for salons, clinics and group classes

Simple

Industry templates: set up the business by answering questions

Transparent

Basic and Pro plans, no per-booking fees

AI-powered

Customers book by chatting with an assistant (Gemini)

OBJECTIVES

General and specific objectives



GENERAL OBJECTIVE

Develop a SaaS booking and appointment platform that is easy to set up for any service business, without the complexity of existing alternatives.

1

✓ ACHIEVED

Manage the project with Jira and Confluence

49 issues tracked across 5 sprints and 19 documentation pages with change control.

2

✓ ACHIEVED

Design a generic business model

A single Service ↔ Resource model covers individual professionals, composite resources and group services.

3

✓ ACHIEVED

Bring AI into scheduling

A Gemini assistant that checks availability, recommends a professional and completes the booking.

SCOPE

Project scope and limitations

In scope (MVP)

- **7 modules** IAM, Tenant, Catalog, Booking, Notifications, Reporting and AI Assistant
- **2 pilot verticals** Beauty salon and medical clinic or practice
- **4 roles** Super Admin, Business owner, Staff and End customer
- **Basic and Pro plans** The AI assistant is available on the Pro plan only
- **Public landing page** Free 14-day demo request

Limitations

- No online payments or public marketplace
- Email-only notifications (no WhatsApp or SMS)
- Responsive web, no native mobile app
- No password recovery (post-MVP)
- End customers book without an account
- Gyms supported by the model, not demonstrated
- No AWS deployment: demo in containers and on video

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METHODOLOGY

Work methodology: Scrum with full traceability

01

Backlog

As a / I want / So that stories, MoSCoW and Fibonacci story points

02

Planning

~2-week sprints with defined acceptance criteria

03

Development

Commits on develop tagged with the Jira story key

04

Review

Pull Request to main, peer review and green CI

05

Done

Story moved to “Done” and documented in Confluence

Jira

Backlog, sprints and board · CAPDC project

Confluence

Living documentation · EXPERTIA-CAP26 space

GitHub

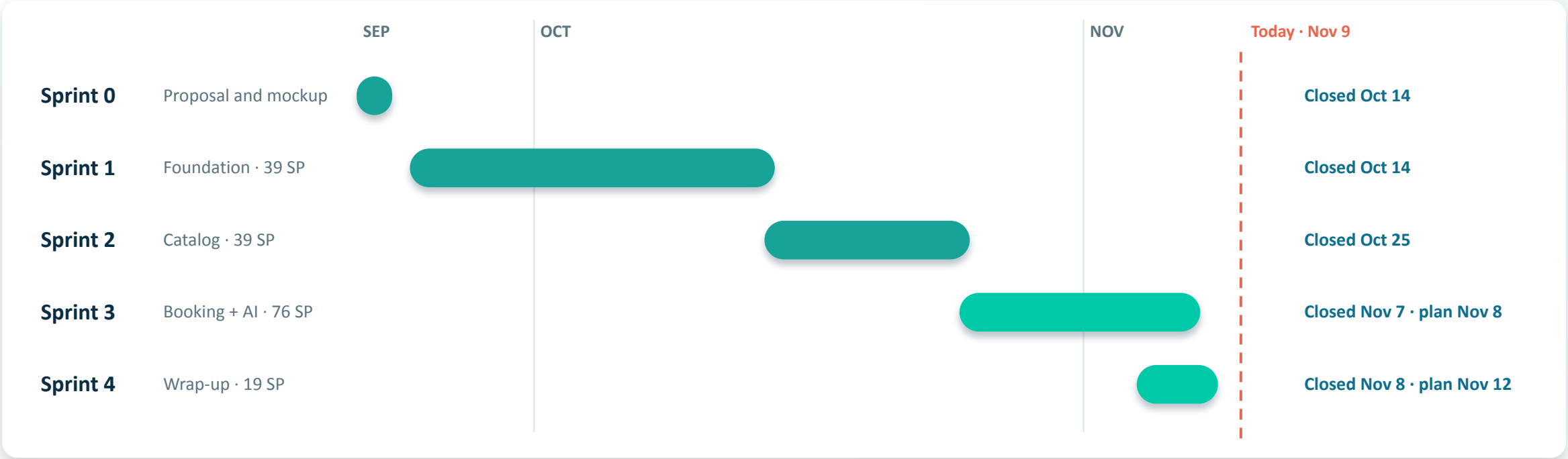
Code, Pull Requests and GitHub Actions

Sprint 0 + 4 sprints

3-person team, ~2-week sprints

TIMELINE

Schedule and completion at close



100% **completion**
173 of 173 planned SP

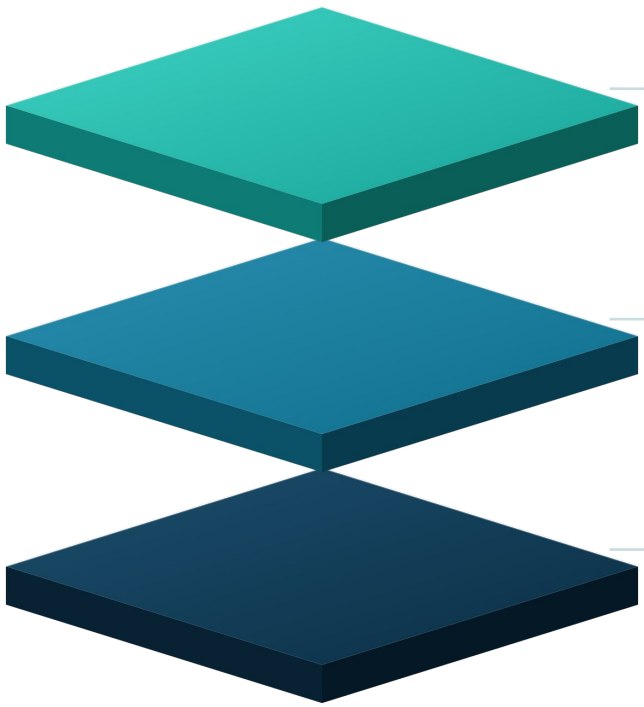
29/29 **stories Done**
All 7 epics complete

48/49 **issues closed**
Pending: Deliverable 3 (CAPDC-47)

4 days **ahead**
Sprint 4: plan Nov 12, closed Nov 8

ARCHITECTURE

Software architecture: modular monolith



FRONTEND

Angular 18 · SPA

BACKEND

Django REST · Python 3.12

DATA

PostgreSQL 16

Everything runs on Docker Compose · REST API with JWT

7 MODULES (BOUNDED CONTEXTS) + SHARED KERNEL

- IAM
- Tenant
- Catalog
- Booking ★
- Notification
- Reporting
- Assistant
- Shared Kernel

LAYERS PER MODULE

- API
- Application
- Domain
- Infrastructure

Assistant powered by Google Gemini Flash

Via LlmProviderPort: the LLM can only call existing Catalog and Booking services (tool calling).

- ADR-001 Modular monolith
- ADR-002 Ports
- ADR-004 Django
- ADR-005 LLM

DATA MODEL

Data model: 13 tables in PostgreSQL

Identity and business

- **tenants**
- usuarios
- miembros_staff

Catalog

- **servicios**
- recursos
- servicio_recurso
- disponibilidad_recurrente
- excepciones_disponibilidad

Booking (core)

- **reservas**
- reserva_recurso

AI Assistant

- **conversaciones**
- mensajes

Notifications

- **notificaciones**

Service ↔ Resource is many-to-many: it enables the generic model (1 resource at a salon; room + equipment + professional at a clinic).

UUID

as primary key

EXCLUDE

GiST: a resource can never be double-booked

tenant_id

isolates each business’s data

BCNF

with one documented denormalization

TECHNOLOGIES

Technologies used



Angular 18

TypeScript SPA frontend



Django REST

REST API · Python 3.12



PostgreSQL 16

pgcrypto and btree_gist



Docker Compose

Full containerized environment



Google Gemini

Flash model for the assistant



JWT

Authentication with simplejwt



GitHub Actions

CI: build, tests and lint



pytest

Backend tests · 80% threshold



Jasmine + Karma

Headless frontend tests



Ruff

Lint and architecture rules (import-linter)



Jira

Scrum, backlog and sprints



Confluence

Living documentation

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RESULTS

Results achieved

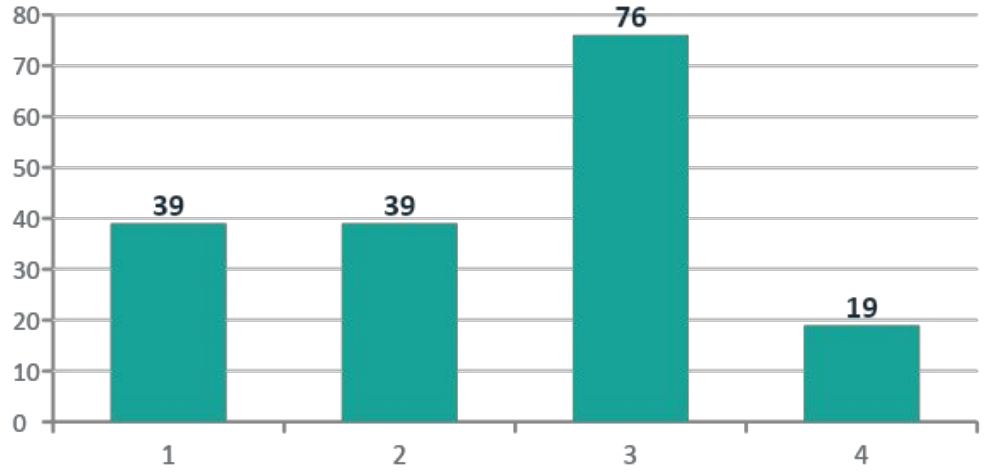
173/173
story points completed

29/29
stories in Done

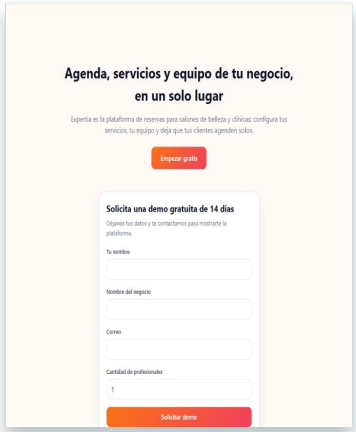
7/7
modules implemented

99.6%
backend coverage measured Oct 7

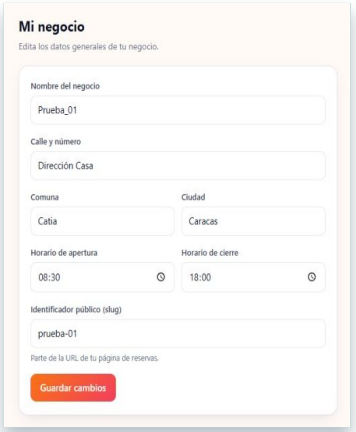
Story points completed per sprint



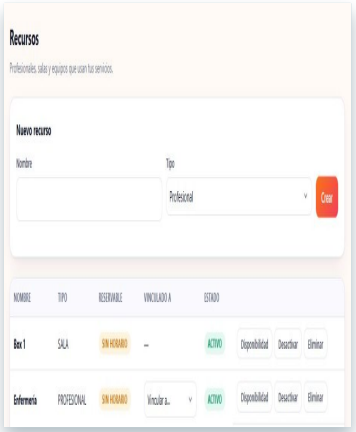
Solution built



Landing and demo



My business



Resources

LESSONS LEARNED

Challenges faced during development

Scope +42%

The AI assistant raised Must-have scope from 92 to 131 SP.

✓ Full Epic 7 delivered in Sprint 3, chatbot kept intact.

Earlier schedule

Sprint 1 moved from Oct 17 to Sep 23 with design stages still open.

✓ Setup, CI and quality became Sprint 1 tasks.

Stack change

Spring Boot was replaced by Django REST (ADR-004).

✓ Documents and architecture rules migrated to Django.

Cloud deployment

On Oct 14, AWS was set aside to prioritize MVP functional scope.

✓ Demo on Docker containers; AWS kept as a future improvement.

Demanding Sprint 3

76 SP (44% of total): booking and assistant in two weeks.

✓ Acceptance criteria set upfront; finished Nov 7.

Jira–Confluence gap

Duplicate keys and an unrecorded Sprint 0.

✓ Change log with traceability by date and version.

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Questions from the panel

and live demo



Landing and sign-up



Catalog and resources



Booking with the assistant



Email confirmation



Dashboard and report

Demo walkthrough · EXPERTIA · Business Management Platform